

DEVELOPMENT GUIDE

Development Guide

This guide provides technical documentation for developers working on the Oro Ticket App.

Table of Contents

- 1. \$1
- 2. \$1
- 3. \$1
- 4. \$1
- 5. \$1
- 6. \$1
- 7. \$1
- 8. \$1

Development Environment Setup

Prerequisites

- \$1: 3.6.0 or higher
- \$1: 3.6.0 or higher
- \$1: Latest stable version
- \$1: 13.0 or higher (macOS only)
- \$1: Latest version
- \$1: Recommended IDE

Environment Setup

- 1. \$1
- 2. \$1
- 3. \$1
- 4. \$1
- 5. \$1

IDE Configuration

Visual Studio Code

- \$1
- Flutter
- Dart
- Awesome Flutter Snippets
- Hive Helper
- \$1

Project Structure

Directory Layout

Module Structure

Each feature module follows MVC pattern:

Code Organization

Naming Conventions

Files and Directories

\$1: \$1, \$1

\$1: \$1, \$1

\$1: \$1, \$1

Classes and Types

Import Organization

Code Formatting

Flutter Analysis

Code Style Guidelines

1. \$1: Maximum 100 characters
2. \$1: 2 spaces (Flutter default)
3. \$1: Same line for classes/functions
4. \$1: Document public APIs

Development Workflow

Feature Development

1. \$1

2. \$1

Create module structure

Implement MVC components

Add unit tests

Update documentation

3. \$1

Run static analysis

Execute tests

Peer review

4. \$1

Module Creation

Using GetX CLI (Recommended)

Manual Module Creation

1. \$1

2. \$1

3. \$1

4. \$1

Data Model Creation

Hive Model Creation

1. \$1

2. \$1

3. \$1

Testing

Unit Testing

Controller Testing

Widget Testing

Integration Testing

Running Tests

Deployment

Build Configuration

Android Build

iOS Build

Environment Configuration

Development Environment

Production Environment

CI/CD Pipeline

GitHub Actions Example

Best Practices

Code Quality

1. \$1

Single Responsibility

Open/Closed

Liskov Substitution

Interface Segregation

Dependency Inversion

2. \$1

Separation of concerns

Dependency injection

Testable code

3. \$1

- Minimize rebuilds
- Optimize list views
- Use const constructors
- Lazy loading

State Management

GetX Best Practices

Error Handling

Security

1. \$1

- JWT token validation
- Request signing
- Rate limiting

2. \$1

- Encrypted local storage
- Secure key management
- Input validation

Troubleshooting

Common Issues

Build Issues

\$1: "Hive adapter not generated"

\$1:

\$1: "Platform not found"

\$1:

Runtime Issues

\$1: "Binding not found"

\$1: Ensure proper binding registration in routes

\$1: "State not updating"

\$1: Use \$1 for reactive variables and \$1 for lists

Performance Issues

\$1: "UI freezing"

\$1: Move heavy operations to background isolates

\$1: "Memory leaks"

\$1: Properly dispose controllers and cancel subscriptions

Debugging

Logging

DevTools

Getting Help

- 1. \$1: <https://flutter.dev/docs>
- 2. \$1: <https://pub.dev/packages/get>
- 3. \$1: <https://docs.hivedb.dev/>
- 4. \$1: Tag with \$1, \$1, \$1

\$1: 1.0
\$1: November 2024
\$1: Flutter Developers
\$1: Oromia Transport Agency Development Team